Y1 Chapter 8 Binomial Expansion

1) Expand and simplify where possible:

a) $(2x-3y)^2$ $4x^2-12xy+9y^2$

b) $(x-y)^3$ $x^3-3x^2y+3xy^2-y^3$

c) $(2+x)^3$ $8+12x+6x^2+x^3$

2) Simplify:

a) $(-2x)^3$ $-8x^3$

b) $(3x)^{-4}$ $\frac{1}{81x^4}$

c) $\left(\frac{2}{5}x\right)^2$ $\frac{4}{25}x^2$

d) $\left(\frac{1}{3}x\right)^{-3}$ $\frac{27}{x^3}$

3) Simplify using root notation:

a) $(25x)^{\frac{1}{2}}$ $5\sqrt{x}$

b) $(64x)^{-\frac{2}{3}}$ $\frac{1}{16\sqrt[3]{x^2}}$

c) $\left(\frac{9}{100}x\right)^{-\frac{1}{2}}$ $\frac{10}{3\sqrt{x}}$

d) $\left(\frac{8}{27}x\right)^{\frac{4}{3}}$ $\frac{16}{81}\sqrt[3]{x^4}$

1: Pascal's Triangle

2: Factorial Notation

3: Binomial Expansion

4: Using expansions for estimation

a) Expand $(a+b)^0$
b) Expand $(a+b)^1$
c) Expand $(a+b)^2$
d) Expand $(a+b)^3$
e) Expand $(a+b)^4$

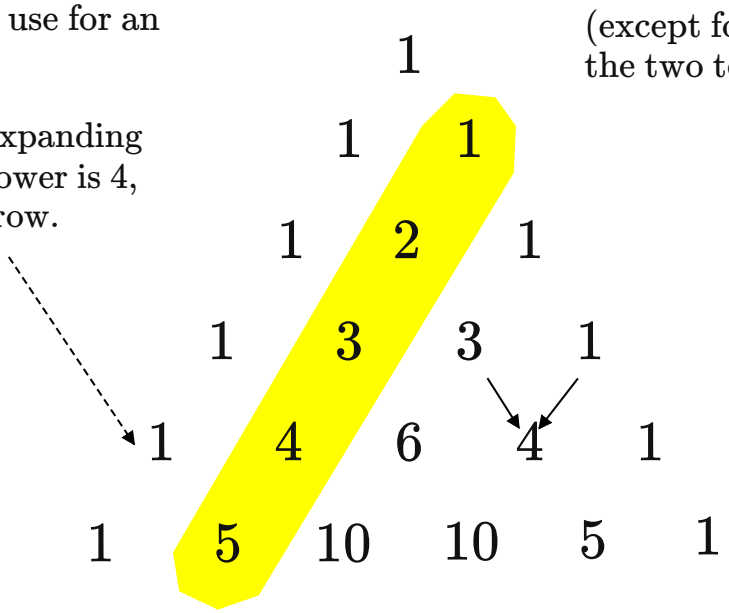
What do you notice about:

The coefficients:
The powers of a and b:
Power of b

The second number of each row tells us what row we should use for an expansion.

So if we were expanding $(2+x)^4$, the power is 4, so we use this row.

In Pascal's Triangle, each term (except for the 1s) is the sum of the two terms above.



Tip: memorise each row up to what you see here.

We'll see later WHY each row gives us the coefficients in an expansion of $(a+b)^n$

Find the expansion of $(2+3x)^4$

Next have descending or ascending powers of one of the terms, going between 0 and 4 (note that if the power is 0, the term is 1, so we need not write it).

First fill in the correct row of Pascal's triangle.

$$
\begin{aligned}
(2+3x)^4 = \quad & 1 \quad (2^4) \\
& +4 \quad (2^3) \quad (3x)^1 \\
& +6 \quad (2^2) \quad (3x)^2 \\
& +4 \quad (2^1) \quad (3x)^3 \\
& +1 \qquad\quad (3x)^4
\end{aligned}
$$

And do the same with the second term but with powers going the opposite way, noting again that the 'power of 0' term does not appear.

Simplify each term (ensuring any number in a bracket is raised to the appropriate power)

$$= 16 + 96x + 216x^2 + 216x^3 + 81x^4$$

Tip: Initially write *one line per term* for your expansion (before you simplify at the end), as we have done above. There will be less faffing trying to ensure you have enough space for each term

$(1-2x)$ is the same as $(1+(-2x))$, so we expand as before, but use $-2x$ for the second term.

$$
\begin{aligned}
(1-2x)^3 = \quad & 1 \quad (1^3) & \\
& +3 \quad (1^2) \quad & (-2x)^1 \\
& +3 \quad (1) \quad & (-2x)^2 \\
& +1 \quad & (-2x)^3
\end{aligned}
$$

$$= 1 - 6x + 12x^2 - 8x^3$$

Tip: If one of the terms in the original bracket is negative, the terms in your expansion will oscillate between positive and negative. If they don't (e.g. two consecutive negatives), you've done something wrong!

The coefficient of x^2 in the expansion of $(2-cx)^5$ is 720. Find the possible value(s) of the constant c.

The '5' row in Pascal's triangle is 1 5 10 10 5 1. If we count the 1 as the '0th term', we want the 2nd term, which is 10.

Since we want the x^2 term:

- The power of $(-cx)$ must be **2.**
- The power of 2 must be **3**

Therefore term is: $10(-cx)^2(2^3) = 80c^2x^2$

$\therefore 80c^2 = 720$

$c^2 = 9$

$c = \pm 3$

(a) Find the first 3 terms, in ascending powers of x, of the binomial expansion of
$(2 + kx)^7$
where k is a constant. Give each term in its simplest form. (4)
Given that the coefficient of x^2 is 6 times the coefficient of x,
(b) find the value of k. (2)

4 Fully expand the expression $(1 + 3x)(1 + 2x)^3$.

5 Expand $(2 + y)^3$. Hence or otherwise, write down the expansion of $(2 + x - x^2)^3$ in ascending powers of x.

6 The coefficient of x^2 in the expansion of $(2+ax)^3$ is 54. Find the possible values of the constant a.

7 The coefficient of x^3 in the expansion of $(2-x)(3+bx)^3$ is 45. Find possible values of the constant b.

8 Work out the coefficient of x^2 in the expansion of $(p-2x)^3$. Give your answer in terms of p.

9 After 5 years, the value of an investment of \pounds500 at an interest rate of $X\%$ per annum is given by $500\left(1+\frac{X}{100}\right)^5$

Find an approximation for this expression in the form $A+BX+CX^2$, where A, B and C are constants to be found. You can ignore higher powers of X.

[MAT 2009 1J]

1 The number of pairs of positive integers x, y which solve the equation:

$$x^3 + 6x^2y + 12xy^2 + 8y^3 = 2^{30}$$

is:

A) 0
B) 2^6
C) $2^9 - 1$
D) $2^{10} + 2$

✎ $n! = n \times (n-1) \times (n-2) \times \ldots \times 2 \times 1$
said "n factorial", is the number of ways of arranging n objects in a line.

For example, suppose you had three letters, A, B and C, and wanted to arrange them in a line to form a 'word', e.g. ACB or BAC.

- There are 3 choices for the first letter.
- There are then 2 choices left for the second letter.
- There is then only 1 choice left for the last letter.

There are therefore $3 \times 2 \times 1 = 3! = 6$ possible combinations.
Your calculator can calculate a factorial using the x! button.

✎ $nCr = \binom{n}{r} = \frac{n!}{r!(n-r)!}$
said "n choose r", is the number of ways of 'choosing' r things from n, such that the order in our selection does not matter.
These are also known as **binomial coefficients**.

For example, if you a football team captain and need to choose 4 people from amongst 10 in your class, there are $\binom{10}{4} = \frac{10!}{4!6!} = 210$ possible selections.
(Note: the $\binom{10}{4}$ *notation is preferable to* $10C4$*)*
Use the nCr button on your calculator (your calculator input should display "10C4")

Calculate the value of the following. You may use the factorial button, but not the nCr button.

a) $5!$
b) $\binom{5}{3}$
c) $0!$
d) $\binom{20}{1}$
e) $\binom{20}{0}$
f) $\binom{20}{2}$
g) $\binom{20}{18}$

a $\times 1 = 120$

b $= 10$

$0! = 1$, but all will be

c

d

e

f

g

If the power in the binomial expansion is large, e.g. $(x+3)^{20}$, it is no longer practical to go this far down Pascal's triangle. We can instead use the choose function to get numbers from anywhere within the triangle. We'll practise doing this after the next exercise.

1	0th row	$\binom{0}{0}$
1 1	1st row	$\binom{1}{0}$ $\binom{1}{1}$
1 2 1	2nd row	$\binom{2}{0}$ $\binom{2}{1}$ $\binom{2}{2}$
1 3 3 1	3rd row	$\binom{3}{0}$ $\binom{3}{1}$ $\binom{3}{2}$ $\binom{3}{3}$
1 4 6 4 1		$\binom{4}{0}$ $\binom{4}{1}$ $\binom{4}{2}$ $\binom{4}{3}$ $\binom{4}{4}$

Notice: The top number matches the row number. The bottom number goes from 0 and eventually matches the top number. It's easy to see from the symmetry of Pascal's Triangle that $\binom{4}{1} = \binom{4}{3}$ for example.

Textbook Note: The textbook refers to the top row as the "1st row" and the first number in each row as the "1st entry". This might sound sensible, but is against accepted practice: It makes much more sense that the row number matches the number at the top of the binomial coefficient, and the entry number matches the bottom number.
We therefore call the top row the "0th row" and the first entry of each row the "0th entry".
So the kth entry of the nth row of Pascal's Triangle is therefore a nice clean $\binom{n}{k}$, not $\binom{n-1}{k-1}$ as suggested by the textbook.

$$\binom{0}{0}$$

$$\binom{1}{0} \quad \binom{1}{1}$$

$$\binom{2}{0} \quad \binom{2}{1} \quad \binom{2}{2}$$

$$\binom{3}{0} \quad \binom{3}{1} \quad \binom{3}{2} \quad \binom{3}{3}$$

$$\binom{4}{0} \quad \binom{4}{1} \quad \binom{4}{2} \quad \binom{4}{3} \quad \binom{4}{4}$$

Each entry of Pascal's Triangle is the sum of the two above it. Thus for example: $\binom{3}{1} + \binom{3}{2} = \binom{4}{2}$

More generally:

$$\binom{n-1}{k-1} + \binom{n-1}{k} = \binom{n}{k}$$

This is known as Pascal's Rule.

10 Given that $\binom{50}{13} = \dfrac{50!}{13!a!}$, write down the value of a.

11 Given that $\binom{35}{p} = \dfrac{35!}{p!18!}$, write down the value of p.

Challenge

a Work out ${}^{10}C_3$ and ${}^{10}C_7$.

b Work out ${}^{14}C_5$ and ${}^{14}C_9$.

c What do you notice about your answers to parts **a** and **b**?

d Prove that ${}^{n}C_r = {}^{n}C_{n-r}$.

In the previous section we learnt about the 'choose' function and how this related to Pascal's Triangle.

$$1 \quad 5 \quad 10 \quad 10 \quad 5 \quad 1$$

$$\binom{5}{0} \quad \binom{5}{1} \quad \binom{5}{2} \quad \binom{5}{3} \quad \binom{5}{4} \quad \binom{5}{5}$$

Why do rows of Pascal's Triangle give us the coefficients in a Binomial Expansion?

Consider: $(a+b)^5 =$ $(a+b)$ $(a+b)$ $(a+b)$ $(a+b)$ $(a+b)$

One possible selection of terms from each bracket.

Each term of the expansion involves picking one term from each bracket.
How many times will a^3b^2 appear in the expansion?

$\mathbb{N}$ is the set of natural numbers, i.e. positive integers. This formula is only valid for positive integers n. In Year 2 you will see how to deal with fractional/negative n.

✎ The binomial expansion, when $n \in \mathbb{N}$:

$$(a+b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \ldots + \binom{n}{r}a^{n-r}b^r + \ldots + b^n$$

Find the first 4 terms in the expansion of $(3x+1)^{10}$, in ascending powers of x.

Find the first 3 terms in the expansion of $\left(2 - \frac{1}{3}x\right)^7$, in ascending powers of x.

5 Find the first 3 terms, in ascending powers of x, of the binomial expansion of $(2-x)^6$ and simplify each term.

6 Find the first 3 terms, in ascending powers of x, of the binomial expansion of $(3-2x)^5$ giving each term in its simplest form.

7 Find the binomial expansion of $\left(x+\frac{1}{x}\right)^5$ giving each term in its simplest form.

Extension

1 [AEA 2013 Q1a] In the binomial expansion of $\left(1+\frac{12n}{5}x\right)^n$ the coefficients of x^2 and x^3 are equal and non-zero. Find the possible values of n.

Hint: Remember that $\binom{n}{2} = \frac{n(n-1)}{2!}$
Can you similarly simplify $\binom{n}{3}$ using $\binom{n}{r} = \frac{n!}{r!(n-r)!}$?

2 [STEP I 2010 Q5a] By considering the expansion of $(1+x)^n$, where n is a positive integer, or otherwise, show that:

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \ldots + \binom{n}{n} = 2^n$$

$$(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \ldots + \binom{n}{n}x^n$$

reflection: This means that the sum of each row in Pascal's Triangle gives successive powers of 2. Safe!

In the expansion of $(a+b)^n$ the general term is given by $\binom{n}{r}a^{n-r}b^r$

Expression	Power of x in term wanted.	Term in expansion
$(a+x)^{10}$	3	
$(2x-1)^{75}$	50	
$(3-x)^{12}$	7	
$(3x+4)^{16}$	3	

The coefficient of x^4 in the expansion of $(1+qx)^{10}$ is 3360. Find the possible value(s) of the constant q.

Term is:

Therefore:
$\mathbf{210q^4=3360}$
$\mathbf{q^4=16}$
$\mathbf{q=\pm 2}$

In the expansion of $(1+ax)^{10}$, where a is a non-zero constant the coefficient of x^3 is double the coefficient of x^2. Find the value of a.

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5 When $(1-2x)^p$ is expanded, the coefficient of x^2 is 40. Given that $p > 0$, use this information to find:

a the value of the constant p

b the coefficient of x

c the coefficient of x^3

6 a Find the first three terms, in ascending powers of x, of the binomial expansion of $(5 + px)^{30}$, where p is a non-zero constant.
b Given that in this expansion the coefficient of x^2 is 29 times the coefficient of x work out the value of p.

7 a Find the first four terms, in ascending powers of x, of the binomial expansion of $(1 + qx)^{10}$, where q is a non-zero constant.
b Given that in the expansion of $(1 + qx)^{10}$ the coefficient of x^3 is 108 times the coefficient of x, work out the value of q.

8 a Find the first three terms, in ascending powers of x, of the binomial expansion of $(1 + px)^{11}$, where p is a constant.
b The first 3 terms in the same expansion are $1, 77x$ and qx^2, where q is a constant. Find the value of p and the value of q.

9 a Write down the first three terms, in ascending powers of x, of the binomial expansion of $(1 + px)^{15}$, where p is a non-zero constant.
b Given that, in the expansion of $(1 + px)^{15}$, the coefficient of x is $(-q)$ and the coefficient of x^2 is $5q$, find the value of p and the value of q.

10 In the binomial expansion of $(1+x)^{30}$, the coefficients of x^9 and x^{10} are p and q respectively. Find the value of $\frac{q}{p}$.

1 *[MAT 2014 1G]* Let n be a positive integer. The coefficient of x^3y^5 in the expansion of $\left(1+xy+y^2\right)^n$ equals:

A) n
B) 2^n
C) $\binom{n}{3}\binom{n}{5}$
D) $4\binom{n}{4}$
E) $\binom{n}{8}$

Try to imagine n brackets written out. To get x^3y^5, we must have chosen xy from 3 brackets, y^2

2 [STEP I 2013 Q6] By considering the coefficient of x^r in the series for $(1+x)(1+x)^n$, or otherwise, obtain the following relation between binomial coefficients:

$$\binom{n}{r}+\binom{n}{r-1}=\binom{n+1}{r}$$

Noting that $(1+x)(1+x)^n=(1+x)^{n+1}$,

(a) Find the first 4 terms of the binomial expansion, in ascending powers of x, of $\left(1+\frac{x}{4}\right)^8$,
giving each term in its simplest form. (4)
(b) Use your expansion to estimate the value of $(1.025)^8$, giving your answer to 4 decimal places. (3)

Tip: Use your calculator to compare against the exact value of 1.025^8.

a

b

(a) Find the first 4 terms of the expansion of $\left(1+\frac{x}{2}\right)^{10}$ in ascending powers of x, giving each term in its simplest form. (4)
(b) Use your expansion to estimate the value of $(1.005)^{10}$, giving your answer to 5 decimal places. (3)

7 a Find the first 3 terms, in ascending powers of x, of the binomial expansion of $\left(3-\frac{x}{5}\right)^{10}$, giving each term in its simplest form. (4 m ...
b Explain how you would use your expansion to give an estimate for the value of 2.98^{10}. (1 ...

8 a Find the first 4 terms, in ascending powers of x, of the binomial expansion of $(1-3x)^5$. Give each term in its simplest form. (4 ...
b If x is small, so that x^2 and higher powers can be ignored, show that $(1+x)(1-3x)^5 \approx 1-14x$. (2 ...

9 A microchip company models the probability of having no faulty chips on a single production run as:

$\mathrm{P}(\text{no fault}) = (1-p)^n,\ p < 0.001$

where p is the probability of a single chip being faulty, and n being the total number of chips produced.

a) State why the model is restricted to small values of p.

b) Given that $n = 200$, find an approximate expression for $\mathrm{P}(\text{no fault})$ in the form $a + bp + cp^2$.

c) The company wants to achieve a 92% likelihood of having no faulty chips on a production run of 200 chips. Use your answer to part **b** to suggest a maximum value of p for this to be the case.